

**AN ANALYSIS OF HOTEL
BOOKING CANCELLATIONS
USER MANUAL**

TABLE OF CONTENTS

TABLE OF FIGURES	2
1. INTRODUCTION	3
2. DATA PRODUCT AND INSIGHTS	3
2.1 PAGE 01 – CANCELLATION OVERVIEW	3
2.2 PAGE 02 – RISK FACTORS AND BEHAVIORAL INSIGHTS	6
2.3 PAGE 03 – CLUSTER SEGMENTATION	10
3. TARGET USERS	13
4. FUTURE RECOMMENDATIONS	13
5. CONCLUSION.....	13

TABLE OF FIGURES

Figure 1 - Cancellation Overview Dashboard	3
Figure 2 - Total Cancellation Metrics	4
Figure 3 - Cancellation Ratio	4
Figure 4 - Monthly Trend by Hotel Type	5
Figure 5 - Country wise Cancellation Distribution.....	5
Figure 6 - Interactive Insights in Cancellation Overview	6
Figure 7 - Cancellation Risk Factors Dashboard	7
Figure 8 - Lead Time vs Cancellations	7
Figure 9 - Average Daily Rate vs Lead Time	8
Figure 10 - Guest Type vs Cancellations	8
Figure 11 - Total Night Stays vs Cancellations	9
Figure 12 - Interactive Insights in Cancellation Risk Factors.....	9
Figure 13 - Cluster Segmentation Dashboard.....	10
Figure 14 - Distribution of Clusters	10
Figure 15 - Cancellation Count by Clusters.....	11
Figure 16 - Average Lead Time and ADR Combined Chart	12
Figure 17 - Interactive Insights in Cluster Segmentation	12

1. INTRODUCTION

The Hotel Booking Cancellation Dashboard provides a powerful, visual representation of patterns, risks, and customer behaviours associated with booking cancellations. This data product enhances decision-making for hotel managers and analysts by delivering insights into cancellation trends, risk factors, customer segmentation, and price sensitivity using interactive visual tools. This Power BI dashboard was developed to tell a clear and actionable story: why do hotel cancellations occur, and how can we proactively manage them? With growing pressure on hotels to predict and react to booking behaviours, this dashboard leverages machine learning and segmentation to support smarter decisions.

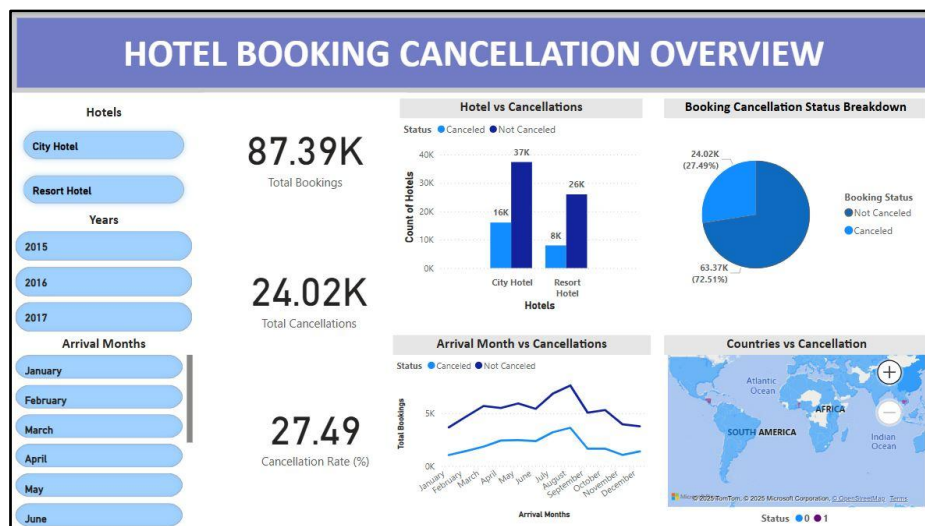
2. DATA PRODUCT AND INSIGHTS

The dashboard includes three major interactive pages, each designed to extract a different layer of insight from the hotel booking dataset:

2.1 PAGE 01 – CANCELLATION OVERVIEW

Figure 1 –

Cancellation Overview Dashboard



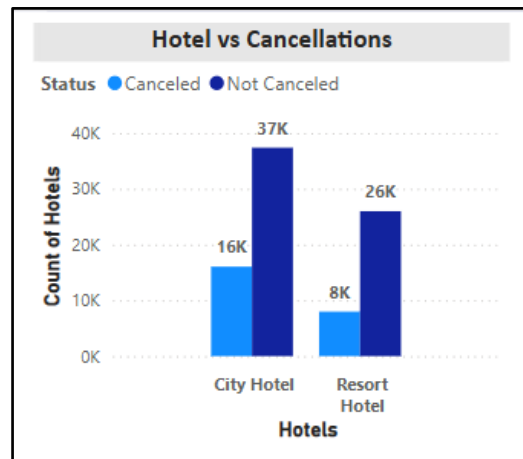
This overview page displays total bookings, cancellations, and their ratio through interactive KPI cards, pie charts, and bar charts. Users can filter by hotel type and month. Geographic insights are provided using a map of cancellations by country.

Key Visualizations :

❑ Total Cancellation Metrics

Figure 2 –

Total Cancellation Metrics

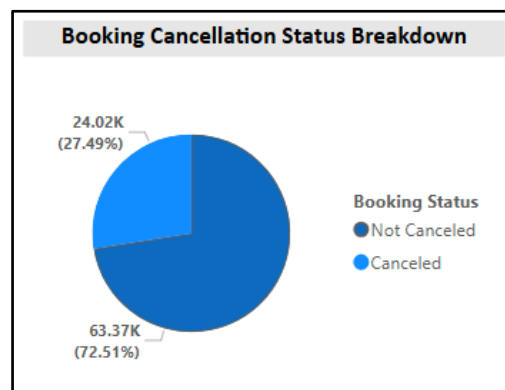


This visual compares cancellations across hotel types and shows that City Hotels experience a higher number of both bookings and cancellations than Resort Hotels. Specifically, City Hotels had 16,000 cancellations compared to 8,000 in Resort Hotels, suggesting that City Hotels may face greater cancellation risk. However, they also have higher overall booking volume, which may explain part of this difference. This insight helps hotel managers understand where cancellations are concentrated and informs people about potential changes to policies or pricing by hotel type.

❑ Cancellation Ratio Pie Chart

Figure 3 –

Cancellation Ratio

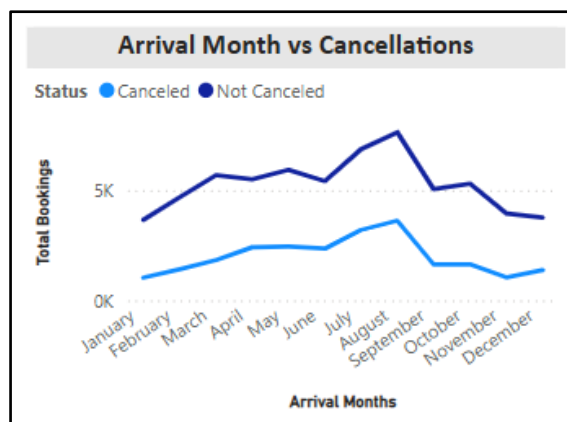


This pie chart shows the overall breakdown of booking statuses, revealing that 27.49% of bookings were canceled while 72.51% were not. With nearly one in four bookings resulting in cancellation, this highlights a substantial area of revenue risk. The chart provides a quick and clear view of the scale of cancellations, helping hotel management understand the impact at a glance and prompting further analysis into why such a high proportion of bookings do not convert to stays.

❑ Monthly Trend by Hotel Type

Figure 4 –

Monthly Trend by Hotel Type



This line chart illustrates monthly booking patterns, showing that cancellations and non-cancellations follow similar seasonal trends, with peaks around July and dips toward December. While non-canceled bookings consistently remain higher, canceled bookings also rise notably during high-demand months like July and August. This suggests that cancellation risk increases during peak travel periods, highlighting the need for better cancellation management and overbooking strategies during those months.

❑ County Wise Cancellation Distribution

Figure 5 –

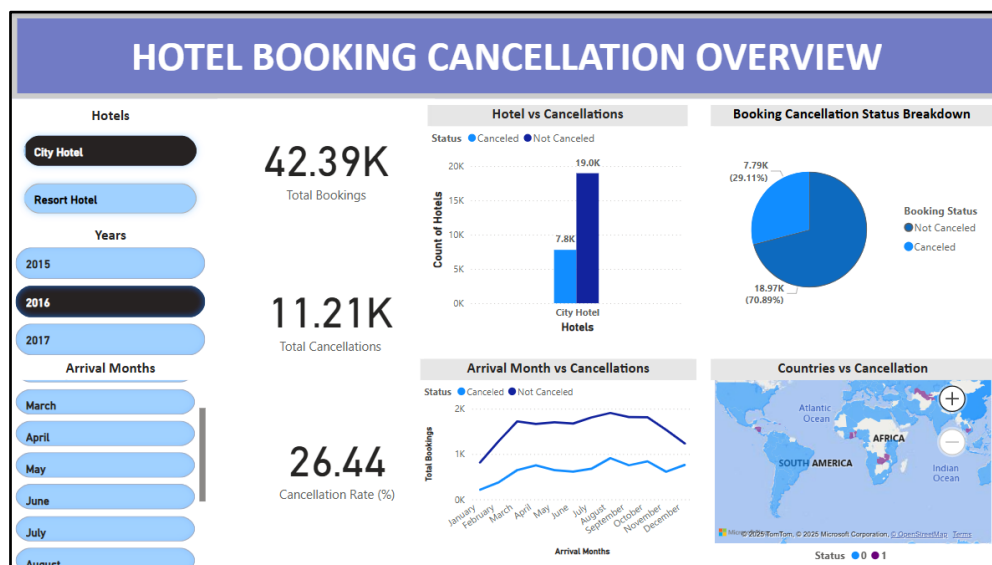
Country wise Cancellation Distribution



This map visualization shows the geographic spread of cancellation and non-cancellation of bookings, where countries that make the most contribution to cancelled and non-cancelled bookings are highlighted. Darker is used to represent greater quantities, with conspicuous clumping in areas such as Western Europe and North America. This information allows for the identification of high-cancellation countries, so hotel managers can target these segments with optimized booking rules or more secure payment options to minimize loss.

Figure 6 –

Interactive Insights in Cancellation Overview



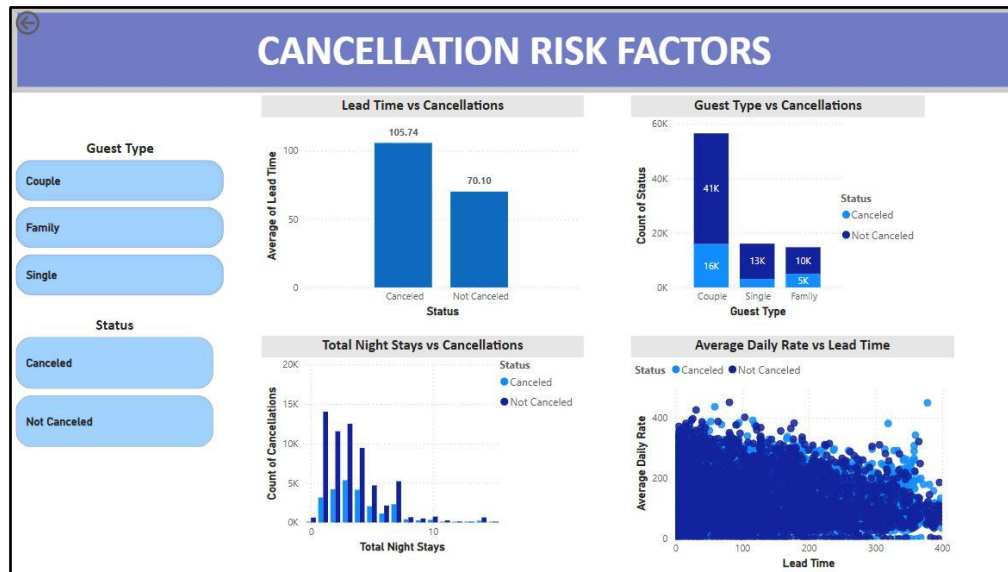
In this interactive dashboard, users can click on any slicer such as hotel type, year, or arrival month to dynamically update all visualizations across the page. This allows users to filter and explore cancellation patterns based on specific conditions, making it easy to compare trends across different time periods, hotel types, or seasonal periods. The charts instantly respond to slicer selections, offering a tailored view of the data and helping users gain deeper, more focused insights.

2.2 PAGE 02 – RISK FACTORS AND BEHAVIORAL INSIGHTS

This page helps stakeholders explore what drives cancellations. It includes scatter plots showing the relationship between lead time and ADR, bar charts by guest type, and insights into how long stays impact cancellation likelihood.

Figure 7 –

Cancellation Risk Factors Dashboard

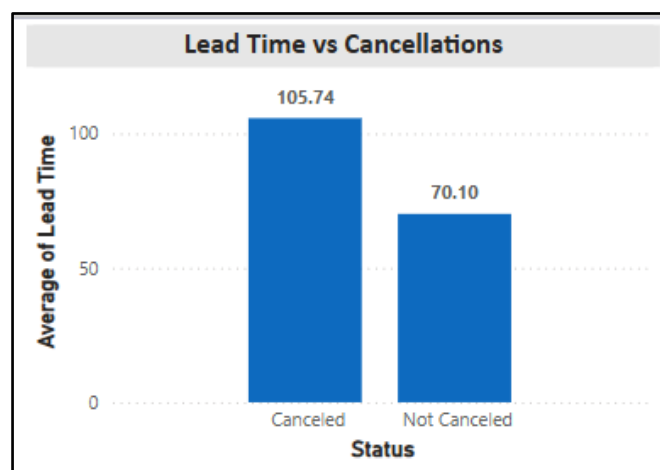


Key Visualizations :

☐ Lead Time vs Cancellations

Figure 8 –

Lead Time vs Cancellations

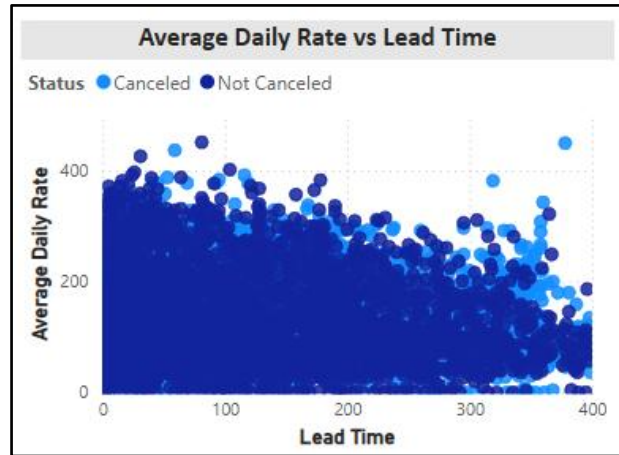


This bar chart shows that cancellations have a significantly longer average lead time. This suggests that earlier booking guests are more likely to cancel, either because of change of plans over time or because of the accommodation nature of cancellation terms. This finding suggests that hotels may need to apply stricter or more relaxed terms for long lead time bookings in order to reduce cancellation risks.

❑ Lead Time vs ADR Scatter Plot

Figure 9 –

Average Daily Rate vs Lead Time

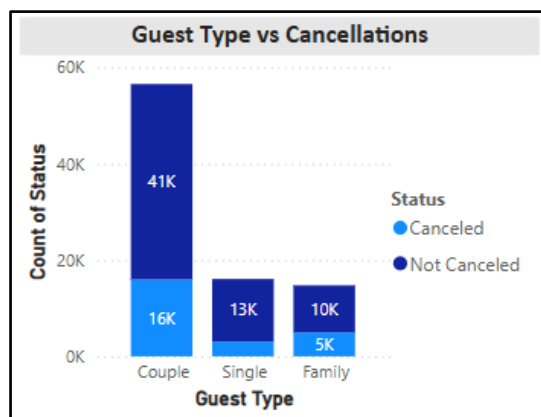


This scatter plot reveals to us that bookings with shorter lead times have more elevated average daily rates (ADRs), and bookings booked well ahead of time have less significant ADRs. Canceled bookings (light blue) are also displayed, which are more concentrated over a wide range of lead times, illustrating that cancellations can occur regardless of how far ahead of time a booking is made. But the direction is towards a pricing curve in which last-minute bookings are at a higher price and their payers are at marginally less risk of cancellation. These facts can help hotels equate pricing strategies with cancellation risk.

❑ Cancellation by Guest Type

Figure 10 –

Guest Type vs Cancellations

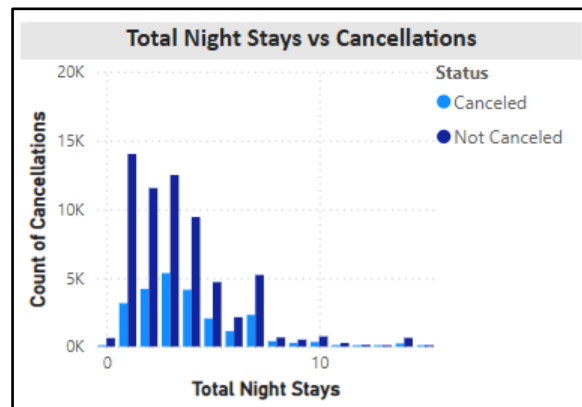


This bar chart shows that couples represent the largest group of both total and canceled bookings. The insight here is that while couples dominate in volume, all guest types contribute to cancellations. This suggests the need for guest-type-specific strategies, such as offering incentives or flexible policies tailored to couples to reduce cancellation rates in this largest segment.

❑ Total Nights vs Cancellations

Figure 11 –

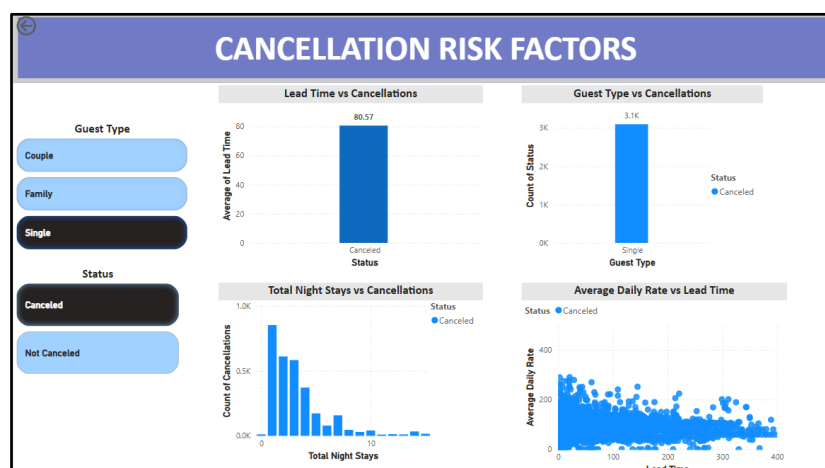
Total Night Stays vs Cancellations



This chart shows that most bookings, both canceled and not canceled, are for shorter stays primarily between 1 and 5 nights. This indicates that short stays make up the majority of hotel business and carry a higher cancellation volume simply due to volume, while longer stays are less common and do not show a strong cancellation trend. This insight can help hotels focus their cancellation strategies more effectively on short-stay bookings.

Figure 12 –

Interactive Insights in Cancellation Risk Factors



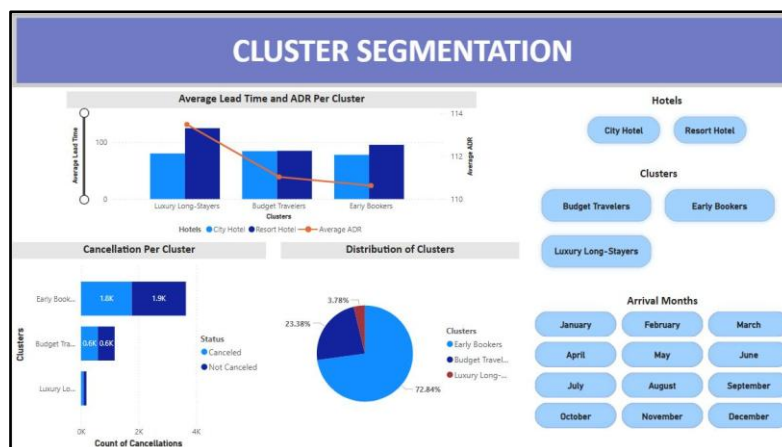
In this interactive dashboard, users can click on the slicers for guest type and booking status to instantly filter and update all the visualizations on the page. This allows users to focus on specific segments, for example, analyzing cancellations only for single guests or comparing lead time trends across canceled vs. not canceled bookings. The dynamic nature of the slicers makes it easy to explore risk factors in a targeted and meaningful way, helping users gain insights tailored to specific booking profiles.

2.3 PAGE 03 – CLUSTER SEGMENTATION

K-means clustering was used to group bookings into customer segments such as early bookers, budget travellers, and long-stay guests. This page visualizes average lead time and ADR per cluster, cancellation counts, and behavioural patterns by month and hotel.

Figure 13 –

Cluster Segmentation Dashboard

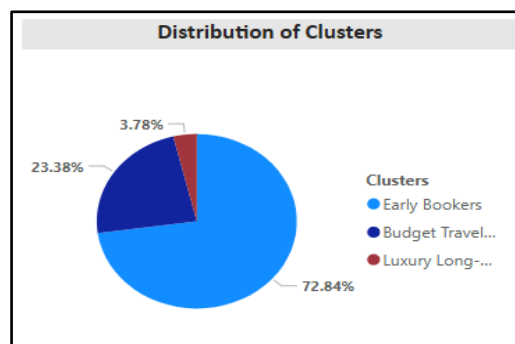


Key Visualizations :

❑ Cluster Distribution Pie Chart

Figure 14 –

Distribution of Clusters

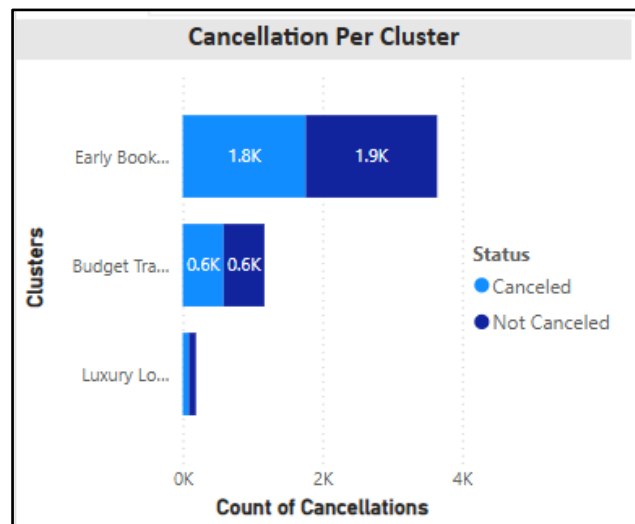


This pie chart demonstrates customer segment composition by grouping, with Early Bookers producing the greatest proportion of reservations (72.84%), followed by Budget Travelers (23.38%), and the least proportion of reservations by Luxury Long-Stayers (3.78%). What this is saying is that most visitors book ahead and upscale long-term customers are a niche market. Having such data about how large each one of these segments is, hotels can more effectively tailor marketing, rates, and cancellation policies to better meet the behavior and expectations of their most loyal customer types.

❑ Cancellation Count by Clusters

Figure 15 –

Cancellation Count by Clusters

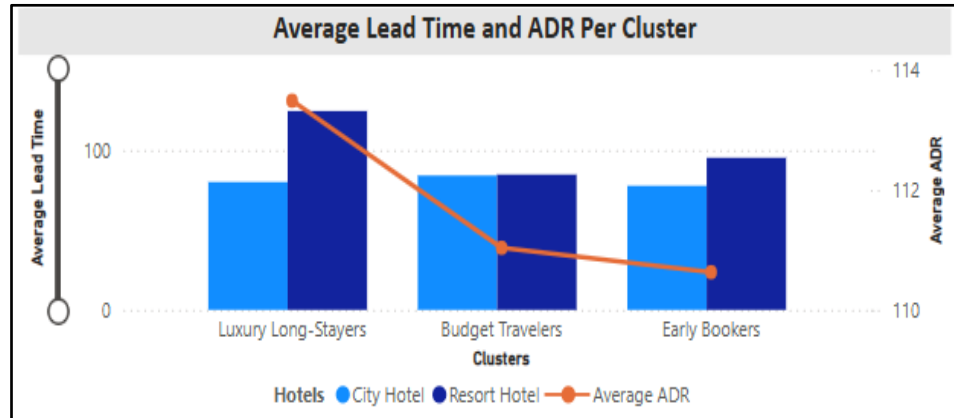


This bar chart compares cancellations by customer segments and shows that Early Bookers have the highest number of cancelled and not cancelled bookings. Budget Travelers have an equal number of canceled and not canceled bookings, suggesting a higher cancellation risk relative to their booking volume. Luxury Long-Stayers make the lowest contribution to overall cancellations, suggesting higher booking stability. These findings can allow hotels to adopt cluster-specific policies, i.e., more restrictive policies for Budget Travelers or early commitment promotions for Early Bookers in order to curtail cancellations.

❑ Average Lead Time and ADR Combined Chart

Figure 16 –

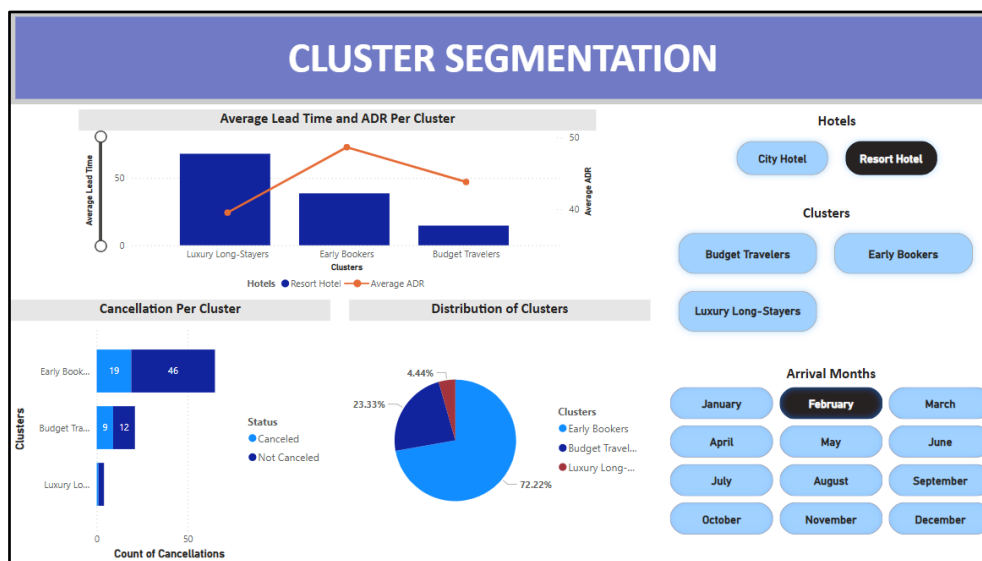
Average Lead Time and ADR Combined Chart



This combined graph shows that Luxury Long-Stayers have the highest average lead time and average daily rate (ADR), especially in Resort Hotels, indicating that they book earlier and spend more. Budget Travelers and Early Bookers have lower lead times and slightly lower ADRs, with similar patterns between hotel types. This discovery allows hotels to distinguish between pricing and promotion strategies for long-stay luxury travelers earlier, while shorter-term promotion management is done for budget and early booking segments.

Figure 17 –

Interactive Insights in Cluster Segmentation



In this cluster segmentation dashboard, users can explore customer behavior by selecting hotel type, cluster, or arrival month from the interactive slicers. All visuals dynamically adjust based

on these selections, allowing users to compare how different customer segments Early Bookers, Budget Travelers, and Luxury Long-Stayers differ in lead time, average daily rate (ADR), and cancellation patterns. This interactive setup enables hotel managers to identify high-value or high-risk segments and make data-driven decisions for pricing, promotions, and booking policies based on targeted customer profiles.

3. TARGET USERS

- ❖ **Hotel Managers:** Use the cluster segmentation and risk visuals to guide inventory management and pricing adjustments.
- ❖ **Revenue Analysts:** Explore lead time and ADR patterns to inform overbooking strategies and price models.
- ❖ **Executives:** Leverage cancellation trends and country-wise performance for strategic marketing and service delivery decisions.

4. FUTURE RECOMMENDATIONS

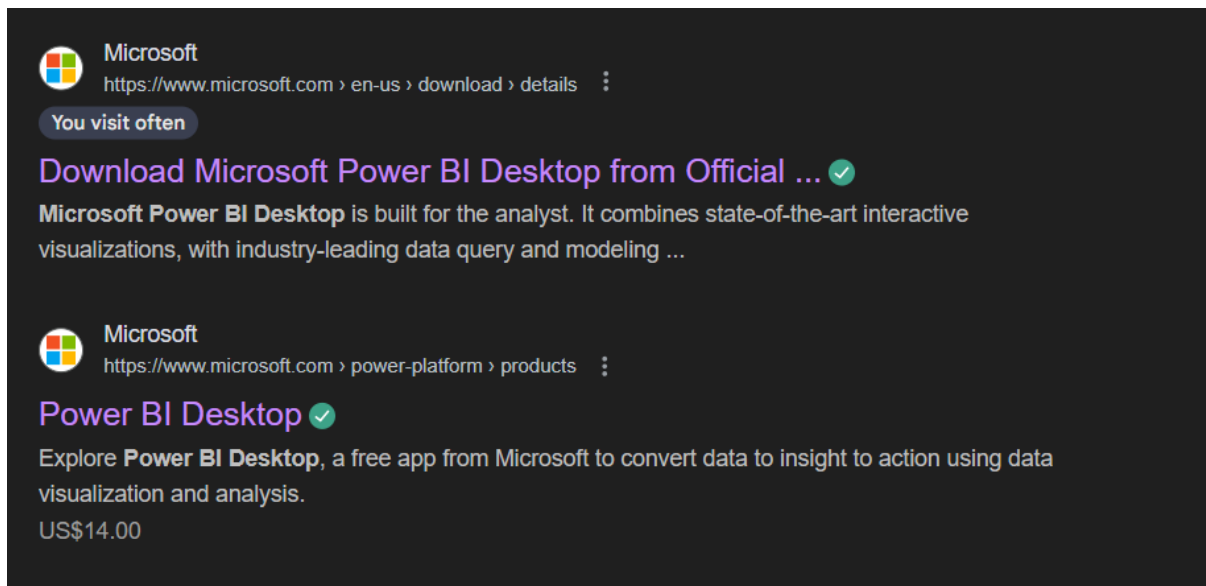
While the dashboard offers a comprehensive view of historical booking behaviour, future improvements could include real-time integration with hotel booking platforms and dynamic forecasting tools. Adding predictive time series models and anomaly detection (e.g., sudden spikes in cancellations) would further support proactive planning and promotional strategies. A role-based dashboard layout (for operations, finance, and marketing) may also improve usability.

5. CONCLUSION

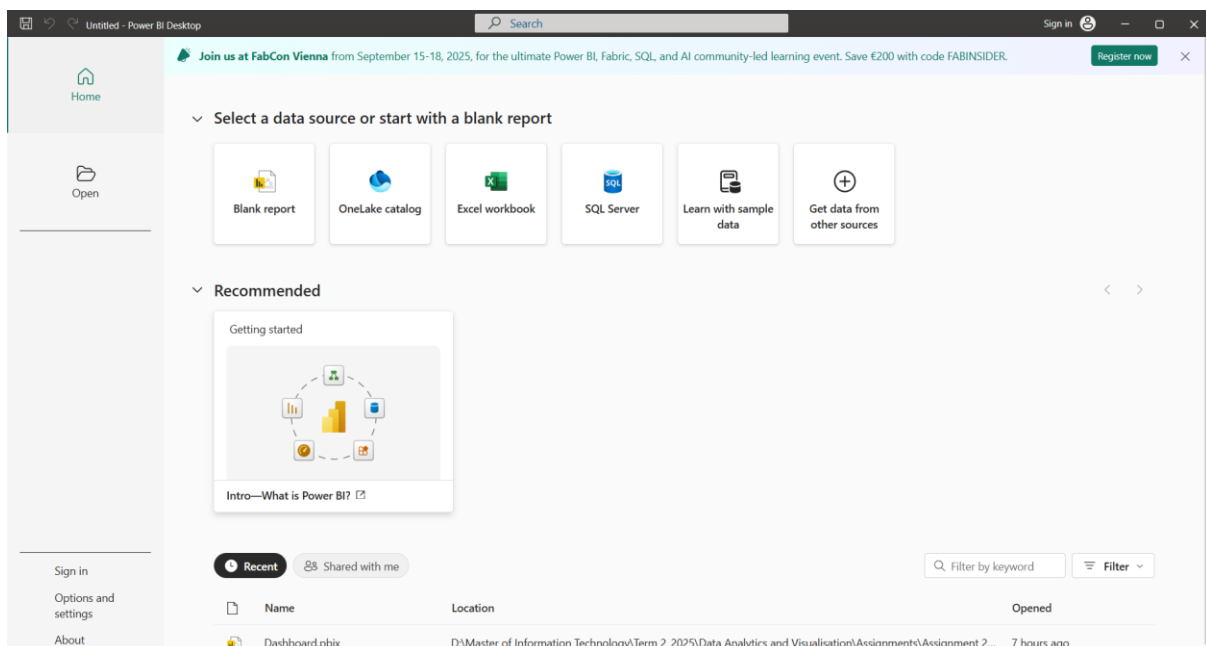
The Hotel Booking Cancellation Dashboard empowers hospitality stakeholders to respond proactively to cancellation risks, optimize resource allocation, and personalize customer strategies. Through its integration of ML model outputs and clustering analysis, it goes beyond static reporting to deliver actionable intelligence. With enhancements in forecasting and real-time data integration, this dashboard can serve as a core decision-support system for modern hotel management.

APPENDIX

Guidelines how to install the Power BI Desktop into OS.



1. Click this to <https://www.microsoft.com/en-us/download/details.aspx?id=58494> open the download option.
2. Then according to your OS you can choose the most compatible exe and then directly download it.
3. After download the exe file you can install the Power BI in your computer.



4. Now you can export your data file to the Power BI and can start to create interaction dashboards.